

As mobile service providers install more antennas and base service stations, the demand for those who specialise in telecom design and implementation is bound to rise

Providers use telecom hardware and therefore employ telecom engineers for their network-related operations.

Opportunities In Hardware

Design and implementation: Designing telecom systems equipment, infrastructure, and networks is a career option—the installation of equipment such as switchers and routers at a base station requires proper design in the first place. However, design and manufacturing of original telecom equipment is yet to start off on a large scale in India. Equipment design engineers, therefore, see better prospects abroad.

For telecom infrastructure, the design architect / solution architect / business analyst is required to have the know-how of voice, data, and fibre-optic networks. Telecom infrastructure design is required for proper transmission voice, data, and other cabling solutions over a network.

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Operations and maintenance: The telecom infrastructure requires operational support and maintenance to function at optimal capacity: this domain includes monitoring, troubleshooting, and analysis of the operations that take place at the base service station. Ultimately, the job of those in the operation and maintenance department is to ensure voice can be transmitted from one end to another.

Network: Job functions in wired, wireless, and data networks all fall under the “network” umbrella. Employed in this domain are engineers who specialise in wireless technologies. They are, for example, GSM engineers / specialists, GPRS engineers / specialists, and CDMA engineers / specialists. Specifically, installation, planning, administration, security management, troubleshooting, and testing of networks are what is involved in this domain.

It should be mentioned that freshers are usually absorbed into the industry in networking jobs. Network engineers at times design terrestrial networks and certify them. Vendor-specific certifications are a plus in a network engineer's profile. Of all the network functions, security is an area worthy of mention in terms of opportunities.

Services: Increased competition means telecom service providers must provide quality service keeping operational costs under control. They deploy solutions for service quality management, fault management, and performance management. This domain therefore comprises service desk functions, service assurance, quality of service management, and such.

Opportunities In Software

Software in the telecom industry usually enables signal testing and network monitoring. Real-time operating systems are of particular importance to aspirants.

Revenue and billing services: Those skilled in DBA (Database Analysis) software systems and data warehousing are employed in this sector. They specialise in working on vendor-based applications and software systems such as Oracle, Kenan, and Siebel. The professional here keeps a check on whether customers are properly charged, and prevents fraud or faulty billing.

Value-Added Services (VAS): A fiercely-growing domain, emerging as a separate industry, VAS includes entertainment, commerce, TV, and more—on the mobile. The majority of the opportunities in VAS are towards programming for the mobile platform. WAP (Wireless Access Protocol) specialisation is in particular demand. Programming and scripting languages used in mobile applications include C, Java, Delphi, Visual Basic, Pascal, PERL, SQL and MATLAB should also be mentioned.

Education

The most obvious and advisable path that leads to a career in the telecom sector is via telecommunications engineering. Only students from a science background can opt for telecommunications engineering or diploma courses: to join a telecom engineering course, one needs to have passed the higher secondary or a vocational course that includes physics, maths, and chemistry in the syllabus. These subjects have been made compulsory as the minimum eligibility criterion to appear for entrance tests such as the IIT Joint Entrance Exams.

In all cases, for employment with a telecom company, the first preference is telecom engineers. Next is electronics and communication engineers, and then those with a B. E. or B. Tech. in computer science.

“The industry generally prefers those with a B. E. / B. Tech. either in computer science or electronics. However, leverage is given for candidates with relevant experience and industry certifications,” according to Shinu Karim, ex-employee, Cable & Wireless Ltd. Aspirants should pursue their B. E. or B. Tech. degree in telecommunications only from universities and

Educational Institutions

The Indian Institutes of Technology

Bombay, Delhi, Guwahati, Kanpur, Madras, and Roorkee

The National Institutes of Technology

Durgapur, Hamirpur, Jamshedpur, Karnataka, Rourkela, Silchar, Srinagar, Trichy, and Warangal

Advanced Level Telecom Training Centre, Ghaziabad

Army Institute Of Technology, Pune

Assam Engineering College, Guwahati

Indian Institute of Science, Bangalore

Bangalore Institute of Aeronautical Engineering and Information

Technology, Bangalore

Bharat Ratna Bhim Rao Ambedkar

Institute of Telecom Training,

Jabalpur

Birla Institute of Technology, Mesra

Birla Institute of Technology and

Science, Pilani

BMS College of Engineering,

Bangalore

College of Engineering, Farmagudi,

Goa

Dayanand Sagar College of

Engineering, Bangalore

Dhirubhai Ambani Institute of

Information and Communication

Technology, Gandhinagar

Dr. Ambedkar Institute Of

Technology, Bangalore

Dr. Babasaheb Ambedkar

Technological University, Mumbai

Hughes Communication India Ltd,

Gurgaon

Jadavpur University, Faculty Of

Engineering And Technology,

Kolkata

Symbiosis Institute of Telecom

Management, Pune

Veermata Jijabai Technological

Institute, Mumbai